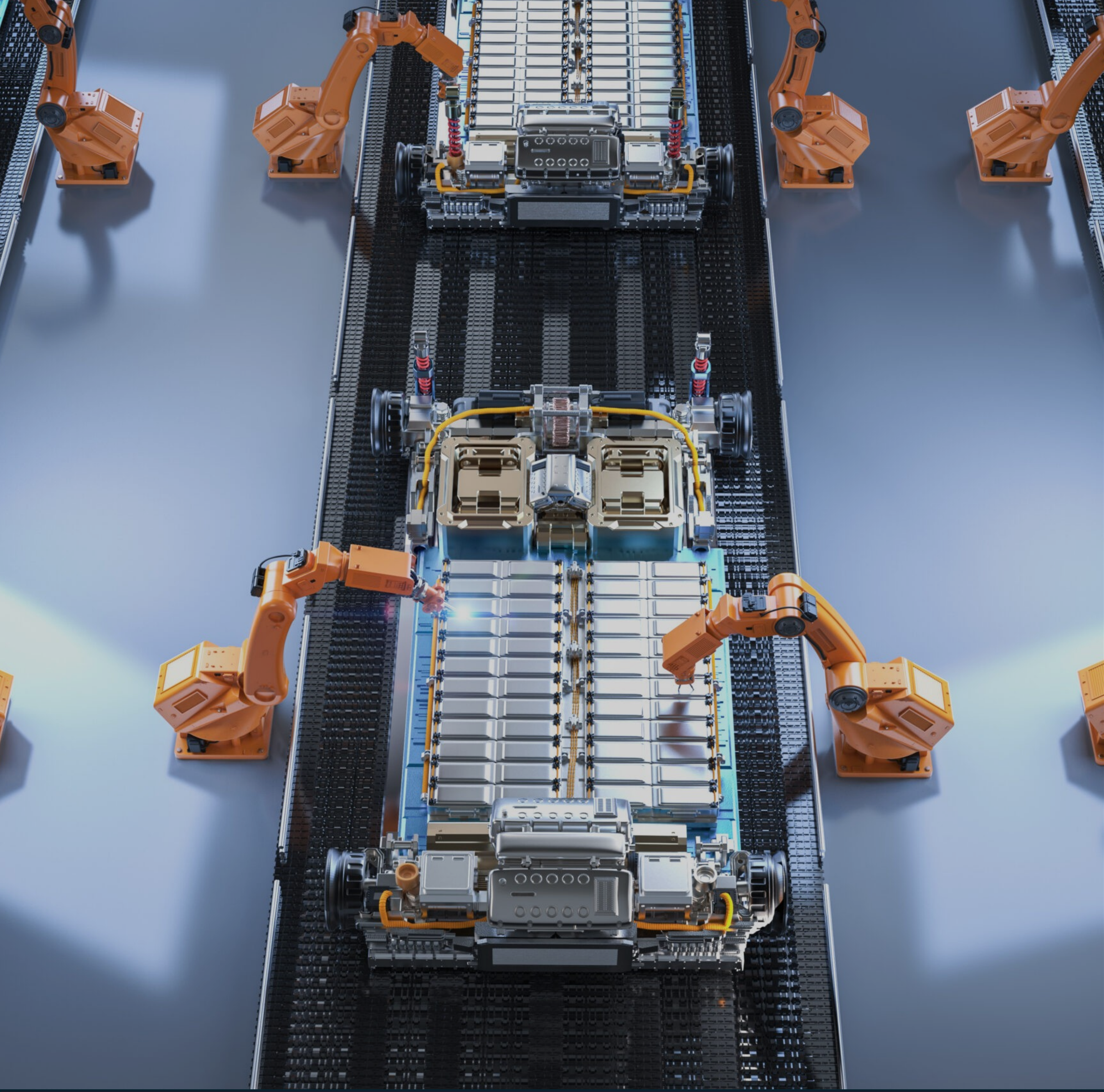

TECHNOLOGY POLICY · STRATEGIC ANALYSIS

SIMON LACEY | MAY 2026

Presence Without Transfer

Manufacturing presence, yes. Technology transfer, no

Market Access X

Presence Without Transfer

How Beijing Is Closing the Technology Exit Window and What It Means for Western Industrial Policy

This report is not investment advice. It is intended for professionals operating at the intersection of trade policy, geopolitics, and strategic supply chain management.

134	65%	77%	Apr 2026
Export control entries, revised TIER Catalogue (2023)	China share of global lithium chemical processing	China share of global battery cell manufacturing	Manus NDRC order: first AI-sector deal publicly blocked

Executive Summary

The NDRC's April 27, 2026 order blocking Meta's acquisition of Manus AI is not an isolated regulatory intervention. It is the public crystallization of a policy architecture Beijing has been constructing since 2021: one that permits Chinese firms to establish manufacturing presence in Western markets while ensuring that process knowledge, engineering talent, and core intellectual property remain domiciled in China.

This report maps that architecture across three sectors -- battery manufacturing, critical minerals processing, and artificial intelligence -- and tests the thesis against evidence that complicates it. The evidence supports a more precise conclusion than the headline suggests: Beijing does not restrict technology transfer categorically. It restricts transfer that exits the Chinese ecosystem. Where Chinese capital retains operational control of the destination -- as in Indonesia's nickel processing complex -- transfer proceeds at scale. Where Western capital seeks to acquire Chinese-origin technology independent of Chinese ownership -- as in the Ford-CATL licensing arrangement now targeted by US legislation, or the Manus acquisition -- Beijing has built or is building a veto architecture. This analysis does not depend on the overcapacity or "China Shock 2.0" framings that have dominated recent Western industrial policy debates. The argument here is narrower and more mechanical: regardless of whether Chinese export expansion is driven by overcapacity, competitiveness, or strategic intent, Beijing has constructed a regulatory architecture that prevents the specific technology transfers Western industrial policy assumes it can access.

The consequence for Western industrial policy is structural. The Inflation Reduction Act, the EU Critical Raw Materials Act, and the proposed Industrial Accelerator Act are all premised, to varying degrees, on the assumption that Chinese manufacturing technology, process knowledge, and engineering talent can be accessed through commercial arrangements while Chinese firms are excluded from the ownership and supply chain positions that would make such transfer commercially rational. Beijing has now formally invalidated that assumption. The window is not closed for every transaction in every sector. But the architecture to close it selectively and permanently is now operational.

Key Findings

- The regulatory architecture governing technology outflows has crossed a qualitative threshold between 2021 and 2026, shifting from a trade-facilitation regime to a security-centric framework.
- The Manus order is the public signal of a policy direction already embedded in export control catalogues, dual-use regulations, and foreign investment security review measures.
- Presence without transfer is the only model Beijing now consistently sanctions for Western partners: physical manufacturing footprint, yes; process knowledge and IP, no.
- Battery manufacturing evidence is strong but nuanced -- CATL's European deployments and Northvolt's collapse illuminate the mechanism, while the Ford-CATL licensing arrangement demonstrates that the restriction architecture is not yet complete.
- Critical minerals processing reveals the deepest structural gap: Western firms cannot replicate Chinese refining economics or process knowledge at commercial scale, and the July 2025 TIER revision has now placed the relevant technologies on the export control list.
- The Singapore incorporation strategy has not failed categorically -- it has been selectively pierced for AI-sector acquisitions and frontier technology exits, while lower-value offshore structures continue to operate.
- The Indonesian HPAL counterexample sharpens rather than undermines the thesis: Beijing transfers when Chinese capital controls the destination supply chain, blocks when it does not.
- Western industrial policy faces a structural premises failure that no amount of subsidy or procurement preference can resolve without confronting the technology access problem directly.

1. The regulatory architecture has crossed a qualitative threshold

The transition is not incremental. The Chinese regulatory environment governing technology outflows has undergone a qualitative transformation since 2021. The defining shift is from a trade-facilitation regime – one that treated technology export as a commercial transaction subject to licensing – to a security-centric framework that treats manufacturing process knowledge as a core national security asset, equivalent in strategic significance to military technology.

The architecture rests on four pillars. The first is the 2023 revision of the Catalogue of Technologies Prohibited or Restricted from Export, issued jointly by the Ministry of Commerce and the Ministry of Science and Technology on December 21, 2023. The revised catalogue contains 134 entries – 24 prohibited, 110 restricted – with new controls added specifically for rare earth extraction, smelting separation, and rare earth magnet manufacturing. The second pillar is the July 15, 2025 revision, which extended controls to battery cathode material preparation technologies, including lithium iron phosphate and lithium manganese iron phosphate, capturing what the regulation defines as fourth-generation LFP at a compaction density threshold of 2.58 grams per cubic centimeter. Crucially, the same revision added controls on the lithium extraction process chain itself: spodumene-to-lithium carbonate, spodumene-to-lithium hydroxide, direct lithium extraction from raw brine, and lithium-containing purified solution preparation. This codified, at the regulatory level, the chokepoint that Western processors had already been hitting commercially.

The third pillar is MOFCOM Decision No. 58 of 2025, effective November 2025, which placed the means of production – winding machines, lamination machines, and liquid injection systems used in battery cell manufacturing – on the dual-use export control list, routing their commercial export through Central Military Commission security review. This move transformed manufacturing equipment into a controlled item, extending Beijing's reach beyond technology licensing into the physical infrastructure of overseas production.

The fourth pillar is the expansion of the NDRC's Foreign Investment Security Review mandate. The 2021 Measures for Security Review of Foreign Investment established the legal framework; the April 27, 2026 Manus order demonstrated its operational reach. The NDRC asserted jurisdiction over a Singapore-incorporated entity on the basis of the Chinese nationality of its founding team and the Chinese origin of its technology – a 'substance over form' interpretation that disregards formal domicile entirely.

REGULATORY ARCHITECTURE: KEY INSTRUMENTS

Dec 2023 -- TIER Catalogue revised: rare earth extraction and smelting separation controlled

Jul 2025 -- TIER updated: 4th-gen LFP, lithium extraction process chain controlled

Nov 2025 -- MOFCOM Decision 58: battery manufacturing equipment on dual-use list

Oct 2025 -- Rare earth export controls (October package); suspended Nov 2025 post-Busan

Apr 2026 -- NDRC Manus order: 'substance over form' jurisdiction over Singapore entity

Apr 2026 -- State Council Provisions on Industrial and Supply Chain Security (Apr 7) and Counter-Extraterritorial Regulations (Apr 13): legal exposure for firms supporting Western de-risking

IMPORTANT: October 2025 rare earth and graphite controls suspended after Trump-Xi Busan meeting -- the architecture is tactical, not absolute

One critical nuance the evidence demands: the architecture is calibrated, not absolute. Graphite export licenses were approved for South Korean battery firms within months of the October 2023 controls, while US and Indian shipments faced two-to-three-month delays. The October 2025 rare earth and graphite controls were suspended in November 2025 as part of bilateral bargaining following the Busan summit. Beijing is building leverage infrastructure – not erecting a permanent wall – which makes the architecture more sophisticated and more durable than a blanket prohibition would be.

A fifth pillar emerged in April 2026 and warrants distinct treatment because it operates on the demand side of the architecture rather than the supply side. The State Council's Provisions on Industrial and Supply Chain Security, issued April 7, 2026, explicitly enable countermeasures against actors engaged in "investigations and other information collection activities" related to China's supply chains, and outline retaliation against companies that suspend "normal transactions" with Chinese firms, apply discriminatory measures, or "cause or threaten substantive harm" to China's industries or supply chains. The Regulations on Countering the Improper Extraterritorial Application of Foreign Laws and Measures, issued April 13, 2026, reinforce this approach by penalizing entities that implement or support foreign "long-arm" jurisdiction deemed improper by Chinese authorities. The first four pillars restrict what exits China. These two regulations create legal exposure for firms that participate in Western de-risking. They are the direct counterweight to the EU's proposed Industrial Accelerator Act conditionality and to US FEOC enforcement. Together, the five pillars do not just police outflows; they raise the cost of inflow restrictions imposed by foreign jurisdictions.

2. The Manus order is the public signal, not the origin point

The NDRC did not invent a new doctrine in April 2026. It applied an existing framework to a new sector with sufficient public visibility to function as a regulatory warning. Understanding the Manus case requires separating the policy signal from the commercial specifics.

The timeline is instructive. Butterfly Effect, the Beijing-based company behind Manus, launched its general-purpose AI agent in March 2025 to international attention. By July 2025, the company had moved its headquarters to Singapore and reduced its Beijing team by approximately 70 percent – a classic offshore restructuring designed to present a foreign identity. Meta announced its acquisition for approximately two billion dollars in December 2025. The Ministry of Commerce initiated a compliance review in January 2026; the NDRC summoned co-founders Xiao Hong and Ji Yichao to Beijing in March 2026 and imposed international exit restrictions. The formal prohibition order followed on April 27, 2026.

The CCTV commentary published after the order made the policy logic explicit. What was being prohibited, state media stated, was 'the non-compliant practice of companies going offshore through washing' – the model by which Chinese-origin technology and talent restructures itself as a foreign entity to avoid Chinese regulatory obligations before selling to Western acquirers. The commentary further argued that because Manus's early-stage R&D was conducted in China by Chinese engineers, 'the movement of its personnel, technology and data is inherently linked to China's interests,' regardless of Singapore incorporation.

This reasoning has direct implications beyond AI. Any Chinese-origin technology company that has undergone offshore restructuring – or is contemplating it – now faces a materially different regulatory environment. The NDRC has demonstrated its willingness to assert extraterritorial jurisdiction based on the substance of a company's technological origins rather than its formal legal domicile. The precedent is most visible in the AI sector, where the Manus order produced the first publicly disclosed prohibition under the 2021 measures, but the underlying doctrine of substance-over-form jurisdiction has already been applied through softer instruments in semiconductors, critical minerals, and strategic infrastructure. The enforcement pattern through softer instruments is independently observable. Beijing reportedly curtailed the transfer of Chinese workforce and equipment to Foxconn’s Indian assembly operations in January 2025, disrupting iPhone production lines. The Ministry of Commerce reportedly delayed approval of BYD’s proposed Mexican EV plant in March 2025 over concerns that proprietary technology could spill over to the United States; BYD subsequently cancelled the project. Geely and BYD have reportedly faced extended review timelines on outbound auto projects through 2025. US-bound outbound FDI approvals were reportedly paused during heightened trade tensions in August 2025. None of these interventions produced a public prohibition order of the Manus type, but each demonstrates the same instrument operating at lower visibility. The architecture exists to harden those softer applications into formal prohibition orders without requiring new legislation.

The Manus case also clarifies what the Singapore strategy was and was not solving. TikTok's Delaware structure failed because the recommendation algorithm's IP had been designated a restricted technology under the 2020 MOFCOM update, and because the 2017 National Intelligence Law created functional obligations that incorporated entities cannot dissolve. Ant Financial's problems were overwhelmingly domestic – a different failure mode that should not be conflated. The Manus case is the clearest example yet of Beijing closing the specific pathway of offshore restructuring followed by Western acquisition – and doing so publicly, with the explicit intention that other companies receive the message.

3. Presence without transfer is the only model Beijing now sanctions

The operating model Beijing has constructed for Chinese clean-technology firms in Western markets can be described precisely: physical manufacturing presence, yes; transfer of process knowledge, manufacturing execution system software, or core IP, no. This is not an informal commercial preference. It is an architecture enforced through export controls on equipment, regulatory guidance on outbound FDI, and – increasingly – personnel controls.

One clarification on the architecture's relationship to commercial pressure is worth stating explicitly. The regulatory instruments described above do not operate against the grain of Chinese firms' commercial interests. They codify the direction those interests already point. Chinese producer prices have contracted continuously since September 2022, with the producer price index roughly five percent below its January 2022 level by March 2026; industrial net profit margins have fallen to their lowest level in at least a decade; corporate R&D growth has slowed in biotechnology, communications equipment, computer hardware, and specialty chemicals; and the RMB real exchange rate against a broad basket of currencies has depreciated materially since 2022. These conditions, drawn directly from People's Bank of China, National Bureau of Statistics, and Bank for International Settlements data, make offshoring high-value production commercially irrational unless tariff walls force the decision. Chinese firms have direct commercial reasons to retain production at home regardless of regulatory direction. What Beijing's architecture does is convert that commercial gravity into a durable policy floor: even firms that might otherwise choose to localize for market-access reasons now face export-control, equipment, and personnel restrictions that close off the offshoring decision at the regulatory level. The architecture is therefore more durable than a pure command system would be, because it is not fighting commercial logic. It is locking it in.

The battery sector provides the clearest evidence. CATL's European deployments are characterized by the deployment of Chinese workers to construct and commission facilities rather than training local staff to acquire those capabilities, and by explicit union statements that CATL does not intend to transfer its manufacturing know-how. The scale of the knowledge gap this leaves was articulated plainly at the November 26, 2025 groundbreaking of CATL's Zaragoza plant with Stellantis, when David Romeral, director general of CAAR Aragon, told Reuters: 'We don't know this technology, these components - we've never made them before. They're years ahead of us. All we can do is watch and learn'. Spain's industry and tourism minister Jordi Hereu, speaking at the same ceremony, described technology transfer as 'fundamental' to the project, framing it as a condition of the partnership's legitimacy rather than an assumed outcome. At the same event, union representatives reported that CATL had no plans to transfer its manufacturing know-how. The gap between the host government's expectations and CATL's reported intentions surfaces the underlying tension without ambiguity.

The personnel dimension has hardened. Article 7 of MOFCOM Announcement No. 62 of 2025 prohibits Chinese citizens, legal persons, and unincorporated organizations from providing 'any substantive assistance and support' for overseas rare earth mining, smelting and separation, metal smelting, magnetic material manufacturing, and rare earth secondary resource recycling activities without permission. While this provision was included in the October 2025 package that was subsequently suspended after Busan, its existence establishes the regulatory instrument. Beijing has signaled that it can extend personnel controls to process engineers, the human carriers of tacit manufacturing knowledge, not merely to equipment and formal IP. The instrument mirrors, with notable symmetry, the restriction the Biden administration imposed in October 2022 prohibiting US persons from supporting China's advanced semiconductor manufacturing without a license, the most aggressive extraterritorial personnel control deployed by Washington in the technology competition to that point. Beijing has now placed an equivalent instrument on the shelf, applicable to the sectors where its own process knowledge advantage is most concentrated.

The Industrial Accelerator Act, proposed by the European Commission in March 2026, attempts to break this model by conditioning market access for non-EU companies with dominant global share on 'real technology and real intellectual property transfer,' with a proposed equity cap of 49 percent for foreign partners in designated sectors. Beijing's response has been to characterize the conditionality as discriminatory and to signal retaliation. The bind is structural: Europe is demanding the transfer that Beijing's architecture is designed to prevent, and the Chinese firms caught between the two sets of requirements have no compliant resolution available under either jurisdiction. Battery manufacturing is where this tension has surfaced most visibly, but the logic applies across every sector where Chinese firms hold the process knowledge that Western industrial policy assumes it can access.

CRITICAL GAP

No publicly available document confirms a formal MOFCOM 'cease and desist' order to any specific Chinese firm regarding European technology transfer obligations. General guidance exists; firm-specific enforcement of the outbound restriction against European IAA conditionality has not been publicly documented. The architecture is nonetheless already shaping commercial behavior.

4. Battery manufacturing: what CATL and Northvolt actually prove

The battery sector evidence is strong but requires careful disaggregation. Three cases – the Ford-CATL licensing structure, CATL's European deployments, and Northvolt's collapse – each illuminate a different dimension of the presence-without-transfer model, and one of them is a direct contradiction that must be stated honestly.

he Ford-CATL BlueOval arrangement is the most important and the most complex. Ford's February 2023 8-K states that its wholly-owned Michigan subsidiary will 'manufacture battery cells using LFP battery cell knowledge and services provided by CATL.' Congressional investigations established that Ford engineers were given access only to the vehicle integration phase, while CATL personnel were required to set up and maintain the manufacturing machinery through approximately 2038. Ford CEO Jim Farley acknowledged the company had 'no legal alternative' but to license the IP because it lacked the internal expertise to replicate Chinese LFP chemistry. This is a textbook presence-without-transfer arrangement, but the party enforcing the knowledge boundary is, as of mid-2026, the United States government, not Beijing. The One Big Beautiful Bill Act, enacted July 3, 2025, explicitly names CATL as a Specified Foreign Entity and bars Section 45X, 45Y, and 48E credits to projects licensing critical technology from CATL under contracts exceeding one million dollars.

The structural irony merits acknowledgment. For three decades, Western automakers entered China under joint venture requirements that Beijing designed to extract technology transfer as the price of market access. Those automakers managed that exposure through their own commercial judgment, retaining frontier platforms onshore and licensing older generations to Chinese partners, with no home government restricting what they could or could not share. The position Ford occupies in Michigan is structurally inverted and mechanistically different in a precise way: Ford has no equivalent commercial discretion. The knowledge boundary is not Ford's to draw. It is drawn on one side by Beijing's export control architecture, which restricts what CATL can transfer, and on the other by Washington's FEOC legislation, which restricts what Ford can receive. The firm sits between two state vetoes, neither of which asked for its preference.

CATL's European deployments – the Debrecen plant in Hungary, the Erfurt plant in Germany, and the Zaragoza joint venture with Stellantis in Spain – are proceeding with advanced LFP technology and substantial Chinese workforce deployment. Stellantis's December 2024 press release explicitly cites CATL technology in Zaragoza; the Spanish industry partner's statement quoted above confirms that knowledge transfer to local staff is not planned. CATL sent approximately 2,000 workers from China to construct and commission Zaragoza; union leaders stated explicitly that CATL 'doesn't plan to transfer its know-how.' There is no evidence of Chinese government blocking action on these deployments – Beijing is permitting Chinese firms to establish European presence under conditions that maintain the knowledge differential.

Northvolt's November 2024 bankruptcy filing and March 2025 Swedish insolvency provide the most visceral validation of the process knowledge gap, though the evidence requires careful handling. Northvolt's Skellefteaa plant delivered less than one percent of its 16 GWh design capacity in 2023. The plant purchased equipment from Wuxi Lead, China's leading battery equipment manufacturer, and reporting from Chinese industry sources suggests that the machines were operable only with Wuxi Lead's own Chinese operators present. A Chinese executive was quoted saying: 'We can raise a factory's battery yield to 96% in just four months. Northvolt took four years and only achieved 70%.' This account – sourced from Chinese industry voices and a Swedish analyst – fits the thesis but has not been verified through Northvolt court filings or BMW disclosures. The Bruegel Institute's more cautious framing -- 'reliance on foreign suppliers for critical inputs' as a systemic vulnerability -- is the more defensible characterization. The conclusion that tacit process knowledge is the binding constraint is strongly suggested; deliberate withholding by equipment suppliers is plausible but not proven.

5. Critical minerals processing: the midstream fortress

The deepest structural gap in the Western industrial policy premise is not at the mine or at the battery factory. It is in the midstream – the refining and chemical conversion processes that transform extracted ore into battery-grade materials. This is where Chinese dominance is most complete, where the process knowledge gap is most acute, and where the July 2025 TIER revision has now placed the relevant technologies under export control.

The concentration figures are not disputed. China refines approximately 65 percent of global lithium, 73 percent of cobalt, 68 percent of nickel, and over 90 percent of rare earth elements. IEA data confirms that for 19 of 20 strategic minerals, China is the leading refiner with an average market share of 70 percent. These figures reflect decades of accumulated process learning, patent accumulation – China filed over 25,000 rare earth patents between 1950 and 2018 compared to approximately 10,000 by the United States – and the development of an industrial ecosystem that integrates tailings management, reactant supply, and process engineering in ways that cannot be replicated by building a single refinery in a jurisdiction without that ecosystem.

The Western experience at commercial scale confirms the gap. Albemarle's Kemerton lithium hydroxide plant in Western Australia was idled in early 2026 after CEO Kent Masters explicitly identified a structural cost gap of four to five dollars per kilogram compared to Chinese facilities. Masters's explanation was precise: 'There's a big industry in China that kind of works through tailings, and we don't have that in the West'. This is not a scale problem that can be solved by building bigger plants. It is an ecosystem problem that requires either replicating the Chinese industrial cluster – a multi-decade project – or accessing Chinese process knowledge through commercial arrangements. Tianqi's Kwinana lithium hydroxide facility, a joint venture with Australian miner IGO, did achieve battery-grade production in 2022 but terminated its Phase 2 expansion in January 2025, citing economic unviability after cumulative preliminary expenditure of approximately 1.4 billion renminbi.

The July 2025 TIER revision codified this vulnerability as a control target. By placing spodumene-to-lithium carbonate, spodumene-to-lithium hydroxide, direct lithium extraction from raw brine, and lithium-containing purified solution preparation on the restricted export list, Beijing has converted a commercial advantage into a regulatory instrument. Western firms that were already failing to replicate Chinese processing economics now face the additional constraint that the technology required to close the gap has been formally restricted from export.

Rare earth processing is the most extreme case. Western alternatives to Chinese separation capacity remain marginal: MP Materials at Mountain Pass, Lynas in Malaysia and its planned Texas and Louisiana facilities, Iluka at Eneabba, Solvay restarting La Rochelle. None has demonstrated commercial-scale heavy rare earth separation capability independent of Chinese process inputs. After China's April 2025 rare earth export controls, European prices reached up to six times Chinese domestic prices – a spread that makes any downstream application of Chinese-separated rare earths in Western manufacturing uncompetitive by definition.

6. The Singapore strategy: selectively pierced, not categorically closed

The Manus case does not establish that offshore incorporation uniformly fails as a regulatory strategy for Chinese-origin technology companies. It establishes that Beijing will pierce the structure selectively, for assets it categorizes as frontier technology with national security implications, when the exit pathway involves Western acquisition of Chinese-origin IP independent of ongoing Chinese control.

The evidence for selective rather than categorical closure is substantial. PDD Holdings, operating through a Cayman Islands structure with its principal operating subsidiary in Shanghai, has expanded Temu globally without NDRC intervention – though it now faces an EU Foreign Subsidies Regulation investigation, which is a Western regulatory action rather than a Chinese one. Shein successfully operated as a Singapore-headquartered firm for years; its current complications arise from attempting to satisfy the China Securities Regulatory Commission for a Hong Kong IPO, which requires disclosures about Xinjiang supply chain risk that CSRC has resisted. This is not Beijing closing the Singapore structure – it is Beijing asserting conditions on the capital raising that accompanies it. The Lazada case, operating under Alibaba as a Singapore-headquartered subsidiary, proceeded without NDRC blocking.

The selection criterion that emerges from the evidence is consistent: Beijing pierces the structure when the transaction would transfer frontier technology out of the Chinese ecosystem to a Western acquirer who would then own and control it independently. The Manus case triggered this criterion clearly – an AI agent architecture built in Beijing, by Chinese engineers, sold to Meta for two billion dollars, with the intention of folding it into Meta AI. The TikTok algorithm case triggered the same criterion through a different mechanism: MOFCOM's 2020 designation of the recommendation algorithm as a restricted technology under the TIER Catalogue meant that any sale or structural separation required export licensing approval that was never going to be granted for a transfer to US ownership.

The practical implication for companies considering offshore restructuring is not that the strategy is dead – it is that the strategy requires genuine separation of Chinese-origin technology from the entity being restructured, which is operationally very difficult for AI and advanced manufacturing companies whose core value is precisely the knowledge embedded in their Chinese-trained engineering teams. A company that relocates its headquarters to Singapore while retaining the same Chinese engineering team working on the same technology has not created the substance that would survive an NDRC 'substance over form' review.

7. The contradiction that sharpens the thesis: Indonesia

The most important piece of evidence in this entire analysis is the one that appears to contradict the thesis. Chinese firms have transferred high-pressure acid leach processing technology to Indonesia at unprecedented scale, with explicit Chinese government approval, faster and cheaper than anything the Western industrial policy community has managed to replicate. This is not a marginal case. It is the largest single technology transfer event in the critical minerals sector in the past decade, and it demands a precise explanation.

The scale of Indonesian HPAL deployment is striking. PT Huayue Nickel Cobalt at the Indonesia Morowali Industrial Park, PT Halmahera on Obi Island, PT QMB New Energy Materials, PT Huafei, PT Huake – these facilities, built through joint ventures involving Tsingshan, Huayou, CATL, GEM, CMOC, and Eve Energy, have commissioned over 160,000 tonnes of nickel-in-MHP capacity, with the Huayue facility exceeding its nameplate capacity of 65,000 tons per year. Wood Mackenzie estimates capital costs of approximately 30,000 dollars per ton of annual nickel capacity in Indonesia, compared to approximately 100,000 dollars elsewhere – a cost differential that makes HPAL in Western jurisdictions commercially unviable at current nickel prices. Bloomberg's 2024 investigation documented thousands of Chinese technical staff on site, 'often taking more specialized engineering and technical roles,' living onsite in conditions that functionally separated them from the Indonesian workforce.

This case does not contradict the thesis. It sharpens it. The selection criterion is not whether the destination is Chinese-owned. It is whether Chinese capital retains operational control of the destination supply chain and whether the technology remains embedded in a Chinese-dominated industrial ecosystem. A second filter operates alongside the ownership test: host-jurisdiction conditionality. Beijing has reportedly instructed Chinese automakers to pause large investments in EU member states that supported tariffs on Chinese EVs, while channelling investment toward politically aligned destinations such as Hungary. The CATL Zaragoza plant occupies the resulting middle position: Chinese-controlled and therefore permitted under the ownership criterion, but located in a jurisdiction where the host government has publicly framed technology transfer as “fundamental” to the project’s legitimacy. The union and local-industry statements about non-transfer of know-how, cited above, are how the contradiction resolves at the operational level — presence proceeds, transfer does not. The refined criterion is therefore two-part: Beijing permits Chinese-controlled overseas presence where (a) Chinese ownership is retained and (b) the host jurisdiction is not deploying conditionality designed to extract process knowledge. Where (b) is breached, the response is not to withdraw but to maintain presence while operationally withholding the knowledge transfer the host expects. Every major Indonesian HPAL facility is majority-owned by Chinese firms. The technology traveled with the capital – but the capital stayed Chinese. The nickel and MHP output from these facilities feeds directly into Chinese battery supply chains: CATL, BYD, LG Energy Solution through Chinese-controlled cathode plants.

Contrast this with the Tianqi Kwinana joint venture in Western Australia, where Tianqi held 51 percent and Australian miner IGO held 49 percent - still majority Chinese-owned, but with an Australian partner holding a meaningful equity stake in a Western jurisdiction. That facility achieved Phase 1 production but terminated Phase 2 expansion as uneconomic. The differential is instructive: Indonesian HPAL, where the destination supply chain is Chinese-controlled, scales at unprecedented speed; Australian lithium hydroxide, where the joint structure involves genuine Western co-ownership, stalls. The pattern is consistent with a deliberate selection criterion rather than a random distribution of commercial outcomes.

The Ford-CATL arrangement reinforces this reading from the other direction. CATL was prepared to license LFP technology to Ford -- a US automaker in a 100 percent US-owned facility -- under a commercial arrangement. There is no evidence of MOFCOM blocking or conditioning this arrangement. The party that has moved to restrict it is the US government, through OBBBA's FEOC provisions. Beijing's architecture targets exits from the Chinese ecosystem; the US architecture targets Chinese technology entering US supply chains under any ownership structure. These are different and in some ways symmetrically opposite restrictions, and they are converging on the same set of commercial transactions from both directions.

8. Western industrial policy faces a structural premises failure

The Inflation Reduction Act, the EU Critical Raw Materials Act, and the proposed Industrial Accelerator Act share a common assumption that the evidence examined here calls directly into question: that Chinese manufacturing technology, process knowledge, and engineering talent can be accessed through commercial arrangements while Chinese firms are simultaneously excluded from the ownership and supply chain positions that make technology transfer commercially rational for the Chinese side.

This assumption is not merely optimistic. It is structurally incoherent. Technology transfer from Chinese firms to Western partners has historically occurred through three mechanisms: joint venture structures in which Chinese partners held meaningful equity and operational roles; technology licensing arrangements in which Chinese firms received ongoing royalty revenue and maintained service relationships that preserved their knowledge advantage; and greenfield investments in which Chinese firms built and operated facilities in host countries, deploying Chinese engineering teams and retaining process control. Western industrial policy, as currently designed, has moved to restrict or exclude Chinese equity participation, impose FEOC tests on licensing revenue, and demand technology transfer conditions that require Chinese firms to surrender process knowledge that Beijing's export control architecture now formally prohibits them from transferring.

The result is a policy framework that demands an outcome -- Western processing and manufacturing capacity built on Chinese process knowledge -- while systematically removing the commercial mechanisms through which that transfer would have to occur. The Gotion Michigan project is the clearest single illustration: a 2.4 billion dollar investment terminated not because the technology transfer terms were unacceptable but because the political and regulatory environment rendered any Chinese-affiliated battery investment ineligible for the IRA credits that made the economics viable. The technology access problem was never resolved; it was made moot by the project's termination.

Europe's position is differently structured but equally contradictory. The proposed Industrial Accelerator Act would condition market access for companies with dominant global share -- a threshold that explicitly captures Chinese battery, solar, and wind firms -- on 'real technology and real intellectual property transfer' in joint ventures capped at 49 percent foreign equity. Beijing has characterised this as protectionist and signaled retaliation. The practical effect is to demand transfer under conditions -- minority foreign equity, mandatory IP sharing -- that make investment commercially irrational for the Chinese side. Envision Energy's situation in France and Spain is the archetype: European regulators demand cell production, not just assembly; MOFCOM guidance advises keeping cell production in China; the company has no compliant resolution under both regimes simultaneously.

The deeper problem is timeline. Building processing capacity outside China does not require only capital and policy frameworks. It requires engineering talent that has spent decades learning iterative manufacturing systems through accumulated failure and refinement. Albemarle's Kemerton cost gap is not primarily a capital problem -- it reflects the absence of a tailings management ecosystem, reagent supply chains, and process engineering knowledge that Chinese refining clusters have accumulated over thirty years. No IRA credit or CRMA mandate resolves that gap. It requires either accessing Chinese process knowledge through arrangements that Beijing's architecture is now closing, or accepting a decade-plus timeline for developing equivalent capability from scratch -- a timeline that is functionally incompatible with the net-zero commitments against which the IRA and CRMA are calibrated.

THE CORE ANALYTICAL CONCLUSION

Beijing transfers technology when Chinese capital controls the destination supply chain.
Beijing blocks transfer when Western capital seeks to acquire Chinese-origin technology independent of Chinese ownership.
Western industrial policy demands the latter while progressively restricting the former.
The window for resolving this contradiction through commercial arrangements alone has closed.

Forward-Looking Implications

Horizon	Policy / Regulatory	Commercial Implication
Near term (2026)	NDRC Manus precedent tested in follow-on AI M&A cases. EU IAA formal proposal and Chinese retaliation response. OBBBA FEOC guidance operationalised for battery licensing. October 2025 rare earth control suspension expires November 2026.	Chinese-origin AI company M&A pipeline effectively frozen for Western acquirers absent Chinese regulatory approval. Battery licensing structures reviewed against FEOC exposure. European gigafactory negotiations stall on technology transfer conditionality.
Medium term (2027-2028)	IAA technology transfer conditionality either operationalised or negotiated down. MOFCOM guidance on outbound FDI in clean-tech sectors likely formalized. Further TIER Catalogue revisions targeting manufacturing execution system software probable.	Presence-without-transfer becomes the explicit and only available model for Chinese-Western battery and processing partnerships. Western firms that have not begun alternative processing development face competitive disadvantage that cannot be closed within energy transition timelines.
Long term (2029+)	Architecture either hardened into permanent bifurcation or partially bargained down in a broader US-China technology framework. EU-China technology investment treaty negotiations likely if IAA retaliation escalates.	Two parallel supply chains -- Chinese-controlled and Western-controlled -- with limited interoperability. Cost differential between Chinese-ecosystem and ex-China processing persists absent technology access. Western energy transition timelines require revision or acceptance of continued Chinese supply chain dependency.

Actionable Takeaways

01 Reframe the technology access problem before the policy problem.

Most Western industrial policy analysis treats technology access as a financing question: provide enough subsidy and the capacity will be built. The evidence examined here suggests the binding constraint is process knowledge, not capital. Government affairs leads and supply chain directors should be asking their technical teams a different question: which specific process technologies does our strategy assume we can access from Chinese partners, and which of those are now on the TIER restricted list or subject to personnel controls?

02 Audit any commercial structure that involves Chinese process knowledge without Chinese equity.

The NDRC's 'substance over form' doctrine and MOFCOM's guidance on outbound FDI both point in the same direction: structures that separate Chinese IP from Chinese ownership are the specific target of the emerging architecture. Joint ventures in which Chinese partners hold meaningful equity and operational roles are, paradoxically, more durable than pure licensing or service arrangements -- they remain commercially rational for the Chinese side. Pure licensing structures of the Ford-CATL type are now targeted by both US legislation and, implicitly, by Beijing's export control tightening.

03 Do not treat the Manus order as an AI-sector anomaly.

The legal instruments used -- the 2021 Foreign Investment Security Review Measures, the TIER Catalogue, the personnel exit restriction authority -- are sector-agnostic. The Manus order is the first public application to an AI acquisition, but the architecture applies wherever Chinese-origin technology of national security significance is being acquired by a Western buyer through a structure that relocates the IP without genuine substantive separation. Battery manufacturing execution systems, lithium processing technology, and rare earth separation processes all qualify under this description.

04 For investors in Western processing and manufacturing assets: the cost gap is structural, not cyclical.

Albemarle's Kemerton idling and Tianqi's Kwinana Phase 2 termination are not market-cycle events. They reflect a structural cost differential rooted in ecosystem gaps -- tailings management, reagent supply, process engineering depth -- that cannot be closed by commodity price recovery or additional capital deployment. The July 2025 TIER revision has now placed the technology that would be needed to close that gap on the restricted export list. Assets that assumed Chinese technology access as part of their investment thesis require reassessment.

05 Map the dual-restriction convergence for any Chinese-affiliated clean-tech investment.

Transactions in the battery, solar, wind, and critical minerals sectors now face restriction from both directions simultaneously: US FEOC rules targeting Chinese technology in American supply chains, and Beijing's export control and security review architecture targeting Chinese technology exiting Chinese-controlled supply chains. Any transaction that appears viable under one regime needs to be tested against the other. The Gotion Michigan termination is the template for how these converging restrictions interact in practice.

06 Engage the EU Industrial Accelerator Act process with precise technical demands, not general principles.

The IAA's technology transfer conditionality is the most direct Western attempt to break the presence-without-transfer model. Its effectiveness will depend entirely on the specificity of what 'real technology and real intellectual property transfer' means in implementing regulation. Vague conditionality will be met with vague compliance. Industry participants who want the IAA to produce actual knowledge transfer -- rather than token commitments -- need to engage the Commission's implementing guidance process with technically specific demands about process documentation, personnel training requirements, and manufacturing execution system access.

Indicators to Watch

01 Follow-on NDRC foreign investment security review cases in non-AI sectors

The Manus order established the 'substance over form' doctrine publicly. Whether the NDRC applies it to battery manufacturing or critical minerals M&A -- where the same logic applies but the commercial stakes are higher -- will determine the architecture's ultimate scope.

02 November 2026: expiry of the October 2025 rare earth and graphite control suspension

The Busan post-summit suspension demonstrated that Beijing treats its export control architecture as a bargaining instrument. Whether the controls are reinstated, permanently lifted, or modified at the November 2026 expiry date will signal whether the October 2025 package was tactical signaling or a genuine policy direction.

03 EU Industrial Accelerator Act implementing regulation

The gap between the IAA's stated technology transfer conditionality and whatever appears in implementing regulation will reveal whether Brussels has the political will to enforce a demand that Beijing has already characterised as discriminatory and threatened to retaliate against.

04 CATL Zaragoza production ramp and local workforce skill transfer metrics

If CATL's Zaragoza plant reaches full production using predominantly Chinese technical staff with no measurable skill transfer to local workers -- as the union and local industry statements suggest is the plan -- it will provide the clearest real-world demonstration of the presence-without-transfer model at scale in a Western jurisdiction.

05 MOFCOM guidance on outbound FDI in strategic sectors

MOFCOM's existing guidance advising firms to keep cell production inside China has not been formalized as binding regulation. Formal publication of outbound FDI restrictions for battery, solar, or rare earth processing sectors would represent a significant hardening of the architecture and would directly impact European and US greenfield investment strategies.

06 Indonesian HPAL ramp trajectory and Chinese equity retention

The Indonesian case is the most important counterexample to the blanket restriction thesis. If Chinese firms begin divesting equity in Indonesian HPAL facilities -- or if Indonesian government resource nationalism forces Chinese ownership below control thresholds -- and technology transfer subsequently stalls, it would provide direct empirical confirmation of the selection criterion proposed in Finding 7.

07 Rare earth processing capacity outside China: yield and purity data

MP Materials, Lynas, Iluka, and Energy Fuels are the primary Western rare earth processors. Their ability to achieve battery-grade or magnet-grade purity at commercial scale -- and the process inputs required to do so -- will determine whether the technology gap is closing organically or remains dependent on Chinese process knowledge.

Sources and References

Primary regulatory documents: MOFCOM-MOST Announcement No. 57 of 2023 (TIER Catalogue revision, December 21, 2023); MOFCOM July 15, 2025 TIER Catalogue revision (CSET translation); MOFCOM Announcement No. 58 of 2025 (dual-use battery controls, November 2025); MOFCOM Announcements Nos. 55-62 of October 9, 2025 (rare earth export controls); NDRC Measures for Security Review of Foreign Investment, January 18, 2021; MOFCOM November 7, 2025 suspension notice (post-Busan). MOFCOM TIER Catalogue translated by Georgetown CSET (cset.georgetown.edu).

Battery manufacturing: Ford 8-K, February 13, 2023 (SEC EDGAR); House Select Committee on the CCP, letter to Ford on Chinese battery partnership; Reuters, 'China battery maker CATL breaks ground on Spanish plant, bringing its own workers,' November 26, 2025; Stellantis press release, December 10, 2024; Northvolt board statement, March 12, 2025; KrAsia/36Kr Northvolt analysis; Bruegel Institute, 'Northvolt's struggles: a cautionary tale for the EU Clean Industrial Deal'; Norton Rose Fulbright, 'Working Through the FEOC Maze'; Bipartisan Policy Center, OBBBA FEOC analyses.

Critical minerals processing: Tianqi Lithium Hong Kong Stock Exchange announcement, January 23, 2025; Albemarle Q4 2024 earnings call, February 12, 2025; Fastmarkets, 'Ex-China lithium hydroxide production ramp-ups face delays'; IEA Critical Minerals Market Review 2025; CSIS, 'What China's Ban on Rare Earths Processing Technology Exports Means'; Wood Mackenzie Indonesia HPAL analysis; Bloomberg Indonesia nickel reporting, 2024; Carnegie Endowment for International Peace, Indonesian battery minerals analysis.

Singapore/offshore strategy: Reuters, Nikkei Asia, Fortune, Business Standard reporting on Manus NDRC order (April 27-28, 2026); Internet Governance Project, TikTok algorithm export control analysis; CSRC VIE filing requirements (March 31, 2023); CNBC, Axios, Modaes reporting on Shein domicile history.

EU regulatory framework: European Commission Net Zero Industry Act (Regulation 2024/1735); Mayer Brown, Skadden, JD Supra on proposed Industrial Accelerator Act (March 2026); White and Case, Cleary Gottlieb, Morgan Lewis on EU Foreign Subsidies Regulation enforcement; MERICS, 'Keeping value chains at home'; Bruegel, 'A smart European strategy for electric vehicle investment from China,' July 2025.

Legal and compliance: Hankun Law analysis on Technology Import and Export Regulation; KPMG China tax alert on 2024 licensing procedures; Mayer Brown, White and Case, Freshfields on October 2025 export control package; Bipartisan Policy Center on OBBBA FEOC provisions. Comparative industrial policy analysis: Rhodium Group, 'China's Next-Generation Industrial Policy,' prepared for the U.S. Chamber of Commerce, May 11, 2026.